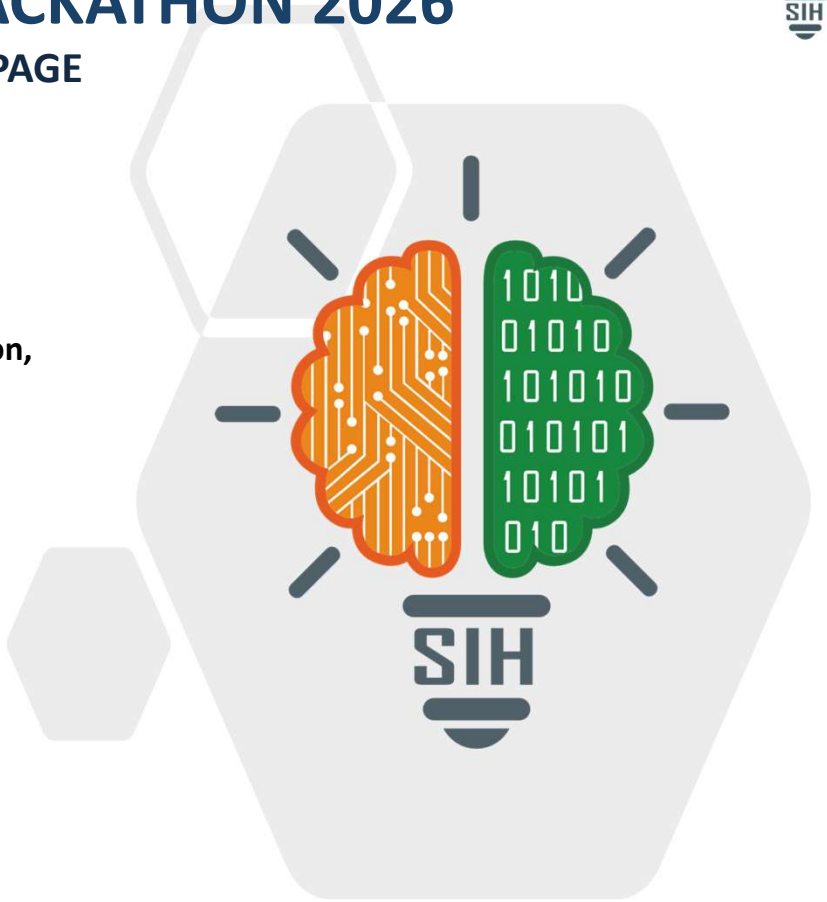
SMART INDIA HACKATHON 2026

TITLE PAGE



- **Problem Statement ID – SIH26261**
- **Problem Statement Title- AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID – 198810**
- **Team Name- AnomalyX**

Team AnomalyX

PRAMAAN-X

PROPOSED SOLUTION

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PRAMAAN-X | AICTE EoA REGULATORY SCRUTINY

Problem: Multi-document EoA dossiers can hide missing evidence, cross-campus conflicts and hard-to-verify records.

WHY SCRUTINY IS HARD

Missing / expired
Mandatory files or dates may be incomplete.

Record conflicts
Faculty, land or facility claims may conflict.

Image alteration / reuse
ELA • copy-move • pHash (Hamming ≤10)

Source verification gaps
Issuer records are simulated; no live APIs.

4-STEP PRAMAAN-X EVIDENCE PIPELINE

01 APP + CHECKLIST
5 statutory PDFs EoA dossier

›

02 PDF FIELD EXTRACTION
PyMuPDF + regex PDF box evidence

›

03 APH RULES + GRAPH
12 APH rules + NetworkX twin

›

04 FORENSICS + ISSUER
ELA / pHash / QR + mock issuer

VALIDATION STATUS

VERIFIED | MISMATCH | EXPIRED | PENDING | SOURCE UNAVAILABLE | MANUAL REVIEW

EVIDENCE-LINKED FINDING
APH clause • PDF box • expected/found • source status

KEY DIFFERENTIATORS

REGULATION-AS-CODE
YAML + JSON Schema; cohort-wide What-If simulation.

IMPOSSIBILITY GRAPH
Ghost faculty, land overlap and temporal impossibilities.

FORENSICS
ELA / pHash / copy-move; QR decode + PDF signature checks.

TAMPER-EVIDENT RECEIPTS
Ed25519 signatures + SHA-256 Merkle chain + DB triggers.

USP: Explainable evidence + impossibility engine + cryptographic receipts. Officer-led; no auto-rejection.

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TECHNICAL APPROACH

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6 CORE MODULES • SYSTEM ARCHITECTURE • SECURITY CONTROLS

3-LAYER TECHNOLOGY STACK

UI / FRONTEND
React 19 • Vite 8 • Tailwind CSS

API + PDF PARSING
FastAPI 0.115.6 • JWT • RBAC

DATA + RULES
SQLite / PostgreSQL • YAML • NetworkX

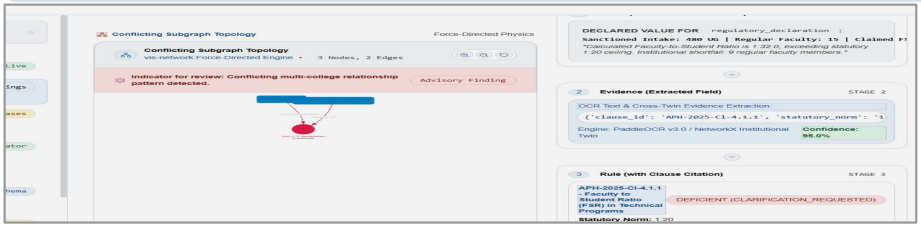
SYSTEM ARCHITECTURE • REACT + VITE → FASTAPI (REST / JSON) • JWT / RBAC / TENANT ISOLATION

1. **CHECKLIST** — EoA + five statutory PDFs
2. **PDF EXTRACTION** — PyMuPDF + regex field boxes
3. **REGULATION-AS-CODE** — 12 YAML rules + What-If
4. **IMPOSSIBILITY GRAPH** — Ghost faculty / land / time
5. **FORENSICS** — ELA • copy-move • pHash QR • PDF ByteRange
6. **VERIFICATION LEDGER** — Ed25519 receipt • SHA-256 chain

DATA: SQLite / PostgreSQL • YAML + JSON Schema • NetworkX twin • append-only DB triggers

DEMO • 3 SERVICES
https://pramaan-frontend.onrender.com
https://pramaan-backend-y3bl.onrender.com
https://pramaan-mock-issuer.onrender.com

EVIDENCE LENS | Claim → PDF box → Rule → Registry status → Officer action



Evidence Lens — Determination View

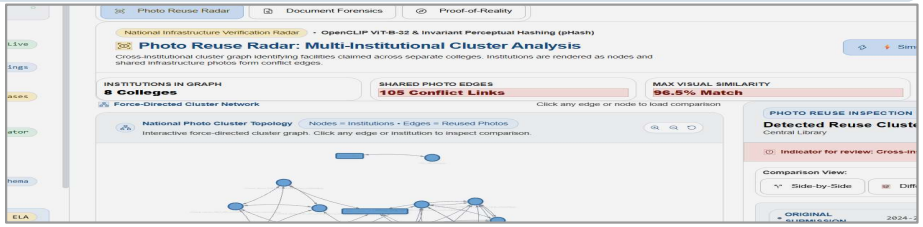


Photo Reuse Radar

BUILT & TESTED • 85 API endpoints • 11 views • 5 RBAC roles • tenant isolation • 12 APH rules • 20 defect types • 182 tests

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FEASIBILITY AND VIABILITY

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FEASIBILITY CHECK

CAPABILITY	MVP APPROACH	CONTROL / FALLBACK
Document AI	PyMuPDF + RapidOCR	Digital PDF text layer; scan OCR; low-confidence → manual review
APH rules	YAML + schema + dry-run	12 YAML rules + JSON Schema + What-If
Graph / forensics	NetworkX • ELA • 64-bit pHash	ELA • pHash • copy-move • QR; X.509 trust-chain stub.
Issuer checks	Fire / land / NBA / affiliation mock issuers	510 mock records; API Setu / DigiLocker not integrated.

RESOURCE & OPERATIONAL VIABILITY

RESOURCE	MVP PLAN	VIABILITY GUARDRAIL
Test data	Synthetic: 100 institutions / 500 PDFs	No production PII • 5 mock roles
Deployment	Local launcher working; Docker Compose config validated (container runtime test pending)	SQLite / PostgreSQL • Docker path
Rule updates	Versioned YAML rule packs	APH sign-off before deployment
Test evidence	182 regression tests passed • 0 failed	15 test files • 182 regression tests

COST & SCALE Open-source • Python 3.12 • React 19 • SQLite / PostgreSQL

KEY RISKS & MITIGATIONS

CHALLENGE 1 • OCR QUALITY & APPROXIMATE SIGNALS

RISK RapidOCR handles scans/images; degraded or handwritten pages need review. OpenCLIP similarity and X.509 trust-chain checks are prototype-level; core pHash / ELA / QR checks are fully implemented.

MITIGATION

✓ Route low-confidence or degraded files to manual review.
✓ Separate real pHash / ELA / QR checks from simulated vector scores.

CHALLENGE 2 • AUTHORITY ACCESS & STATUTORY PARAMETERS

RISK 510 issuer records are synthetic; 12 APH parameters await gazetted sign-off.

MITIGATION

✓ Connect live gateways only after approval.
✓ Mask PAN/Aadhaar; enforce RBAC, tenant isolation and access logs.
✓ Require officer reasons for adverse actions; no auto-rejection.

ROADMAP APH sign-off → authorized pilot → live issuer APIs → multilingual OCR → national rollout with HSM-backed keys.

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IMPACT AND BENEFITS

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HELD-OUT SYNTHETIC BENCHMARK • 100 INSTITUTIONS • 500 DOCUMENTS • NOT FIELD ACCURACY

POTENTIAL IMPACT ON TARGET AUDIENCE

- AICTE officers — use risk-ranked queues and evidence-linked flags.
- Institutions — use the Pre-Flight scorecard and corrective hints.
- Regional teams — prioritize inspections using geospatial risk signals.
- Students & public — benefit from consistent, auditable scrutiny.

BENEFITS OF THE SOLUTION

- Explainability — link claim → PDF box → rule → source response.
- Consistency — prototype covers 20 planted defect categories.
- Auditability — cryptographic receipts and access logs support review.
- Governance — officer action and reasons required; no auto-rejection.

ANOMALY • HELD-OUT SEED 7
240 / 240 planted detected • 0 missed
P 0.9917 • R 1.0000 • F1 0.9959
FPR 3.51% • 2 false flags / 57 controls

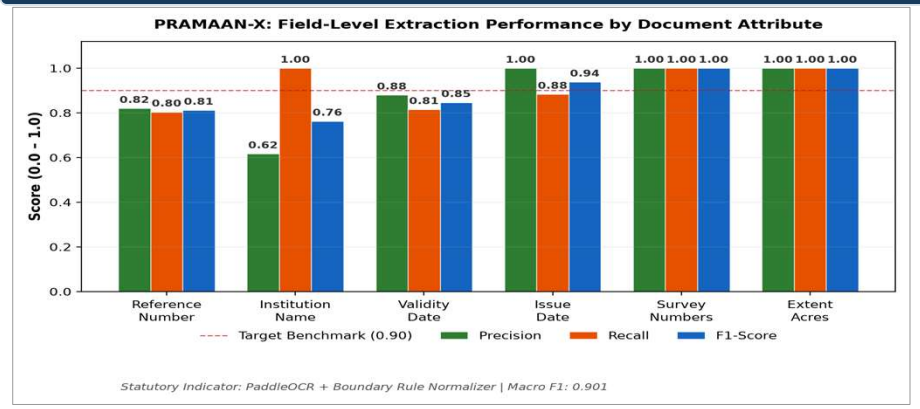
FIELD EXTRACTION • 6 FIELDS
Macro F1 0.9013 • Micro F1 0.8596
TP 199 • FP 38 • FN 27

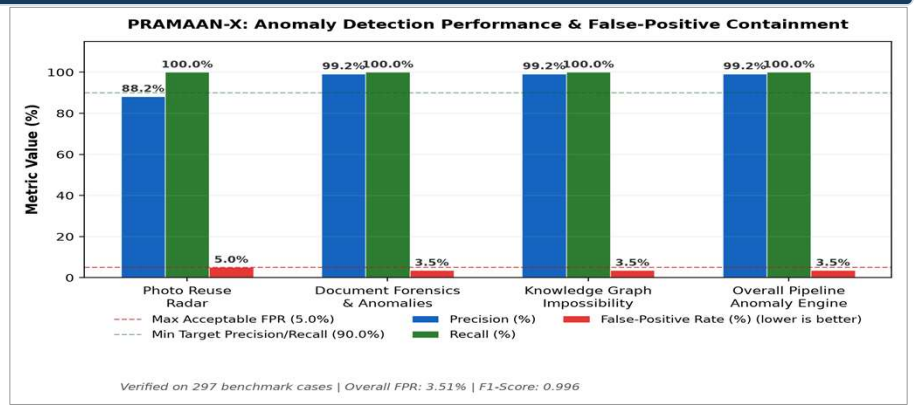
PIPELINE + ALERT LATENCY
2.03 s / dossier
Alerts: 3.21 ms mean • 3.20 ms median

STATUTORY SAFETY / GOVERNANCE
Advisory only • no auto-rejection
5 RBAC roles • tenant isolation • PII masking

Synthetic held-out benchmark only • authority gateways are simulated.

REPOSITORY BENCHMARK CHARTS • HELD-OUT SEED 7





Source: evidence_pack/metrics.json + defect_coverage.csv • synthetic held-out Seed 7 • authority gateways are simulated.

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RESEARCH AND REFERENCES

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KEY REFERENCES • OFFICIAL GUIDANCE + PROJECT DOCUMENTATION

SOURCE	WHY IT MATTERS	PRAMAAN-X USE	SHORT DOMAIN / REPO	SCOPE / NOTE
AICTE APH 2025-26	Approval criteria	YAML rule packs	aicte-india.org	12 engineering placeholders pending gazetted sign-off.
API Setu	Government API platform	Local mock issuer; live API integration not done.	apisetu.gov.in	Live integration not implemented
DigiLocker issuer APIs	Issuer integration guidance	Planned connector target	digilocker.gov.in	Production APIs not connected; approval required.
PyMuPDF + RapidOCR	Digital PDF text + scan OCR	PyMuPDF text layer; RapidOCR for scans/images	pymupdf.readthedocs.io • github.com/RapidAI/RapidOCR	RapidOCR verified on dev; low-confidence cases route to manual review.
NetworkX	Graph analytics	NetworkX digital twin • cross-campus conflicts	networkx.org/docs	Cross-campus signals support officer review.
NIST AI RMF	AI risk governance	Explainability + oversight	nist.gov/AI-RMF	Human accountability • no automated statutory verdict
NIST CSF 2.0	Security guidance	Access / logs / retention	nist.gov/cyberframework	JWT / RBAC • tenant isolation • PII masking / access logs
PRAMAAN-X repository	Evaluation report + metrics	README + metrics + audit	github.com/tejas-coder28/sih-2.0	182 regression tests • synthetic data • 154-element audit report

Academic prototype • synthetic data • mock authority gateways (see slide 4 for full scope).